

STANDARD OPERATING PROCEDURE

KILN HEATER CONTROL

Easy Operator Guide

Purpose

This guide explains how to operate the Kiln Heater safely from the plant dashboard or Cupola widget. It is written for operators and non-technical users.

1. The one rule to remember

The kiln automatic control uses a fixed temperature point of **35°C**. The temperature comes from the kiln temperature sensor.

Temperature	Kiln Heater	Heat Blower	Kiln Motor
Below 35°C	ON	ON	OFF
35°C or above	OFF	OFF	ON

Simple example

If the kiln temperature is 30°C: Heater ON, Blower ON, Kiln Motor OFF.
If the kiln temperature is 36°C: Heater OFF, Blower OFF, Kiln Motor ON.

Important

The dashboard calls this mode “PID Auto-Control”. The current operating rule is the fixed 35°C switching rule shown above. The 35°C value cannot currently be changed from the dashboard.

2. Before you operate

1

Check the system

Make sure the backend and ESP2 device are online.

2

Check the temperature

Confirm the Kiln Temperature value is updating and looks reasonable.

3

Check the mode

Look for “Manual Override Active”. Decide whether you want Automatic or Manual operation.

4

Verify after every command

A message saying “command sent” does not prove the physical relay changed. Check the reported state and follow local plant checks.

3. Automatic operation

1

Open the dashboard

Check ESP2 status and the kiln temperature.

2

Return to Auto if needed

If Master or Manual Override is active, press “Resume Auto PID” on either the Klin or Kiln Heater control.

3

Let the system control the kiln

Below 35°C the heater and blower are requested ON. At 35°C or above they are requested OFF. The kiln motor follows the opposite state.

4

Keep watching

Monitor the temperature and the reported Heater, Blower and Klin states.

Only ONE Resume button is needed

Pressing “Resume Auto PID” on either the Klin control OR the Kiln Heater control clears the Master override and both kiln manual overrides, then immediately checks the latest temperature.

4. Manual kiln heating - follow this order

Why the order matters

For manual heating, the blower must be running before the heater is turned ON. The current software does NOT automatically enforce this sequence.

1

Heater OFF first

Use the Kiln Heater OFF command. This places the heater in manual override and makes sure it starts from OFF.

2

Turn Heat Blower ON

Switch the Heat Blower ON.

3

Verify the blower

Confirm the blower's reported state before continuing.

4

Turn Kiln Heater ON

Only after the blower is ON and verified, turn the Kiln Heater ON.

5

Keep blower ON

Keep the blower running for the whole time the Kiln Heater is ON.

6

Stopping

Turn the Kiln Heater OFF first. Keep the blower running as required by the plant's approved operating procedure.

5. Control priority - which command wins?

MASTER > MANUAL > PID AUTO-CONTROL

MASTER ON / OFF	Controls all 13 relay actuators and pauses kiln automatic temperature control.
Manual ON / OFF	Controls the selected kiln actuator and activates its manual override.
PID Auto-Control	Runs only when Master override is OFF and the relevant manual override is cleared.
Resume Auto PID	Clears Master + both kiln manual overrides and immediately applies the latest temperature rule.

Master control warning

Use MASTER ON / MASTER OFF only when you really intend to command ALL 13 relay actuators. A later individual manual command can change one actuator, but it does not clear the Master pause. Use Resume Auto PID when you intentionally want the kiln branches back in automatic control.

6. Cupola widget

The Cupola kiln-heater widget provides ON and OFF buttons for the kiln heater.

1

Open the widget

Opening the widget page alone does not switch the heater.

2

Select ON or OFF

The command is sent only when the operator selects a button.

3

Verify in the dashboard

Check the reported heater state. The widget does not provide a Resume Auto button.

4

Return to Auto

Use "Resume Auto PID" on either kiln-related dashboard control.

Cupola manual heating

The Cupola heater widget controls only the kiln heater. For manual heating, use the dashboard to make sure the heater is OFF, turn the Heat Blower ON and verify it, then turn the heater ON from Cupola.

7. Preheating Tower Heater - separate equipment

Do not mix these two heaters

The Preheating Tower Heater is separate from the Kiln Heater. Its temperature is currently for monitoring only. The kiln's 35°C automatic rule does NOT control the Preheating Tower Heater.

8. Temperature monitor - easy meaning

Starting Temp	Temperature captured when heater operation starts and no baseline is set.
After Heater Temp	Latest kiln temperature received while the heater is ON.
Temp Gain (ΔT)	After Heater Temp minus Starting Temp.
Reset Baseline	Clears only these monitor values. It does NOT turn the heater OFF and does NOT change 35°C.

Example

Starting Temp = 28°C, After Heater Temp = 33°C → Temp Gain = +5°C.

These monitor values are stored only in backend memory and are cleared when the backend restarts.

9. If something does not work

1

Check ESP2 and backend

If either is offline, commands or automatic control may not operate correctly.

2

Check the temperature sensor

Make sure the kiln temperature is present, plausible and changing.

3

Check Manual Override

If Manual Override is active, automatic control may be paused.

4

Check all three kiln states

Look at klin_heater, heat_blower and klin after temperature updates.

5

Watch around 35°C

Because there is no hysteresis, the system may switch repeatedly if temperature moves above and below 35°C.

10. Safety - read before use

SAFETY LIMITATION

This software is an operating interface, not the only safety protection. It does not provide a validated over-temperature trip, independent safety interlock, or guaranteed fail-safe heater shutdown. Always follow the plant's approved operating, isolation, inspection and emergency procedures. Escalate unexpected physical behavior to the responsible plant operator.

Quick reference

Temperature < 35°C	Heater ON Blower ON Klin OFF
Temperature ≥ 35°C	Heater OFF Blower OFF Klin ON
Need manual heater	Heater OFF → Blower ON → Verify → Heater ON
Return to automatic	Press Resume Auto PID on either kiln control
Master command	Affects all 13 relay actuators and pauses kiln Auto
Preheating tower	Separate heater; no kiln 35°C Auto rule